1 **(a)** Define acceleration.

.....

..... [1]

(b) A diver stands on a platform above a pool of water, as shown in Fig. 1.1.

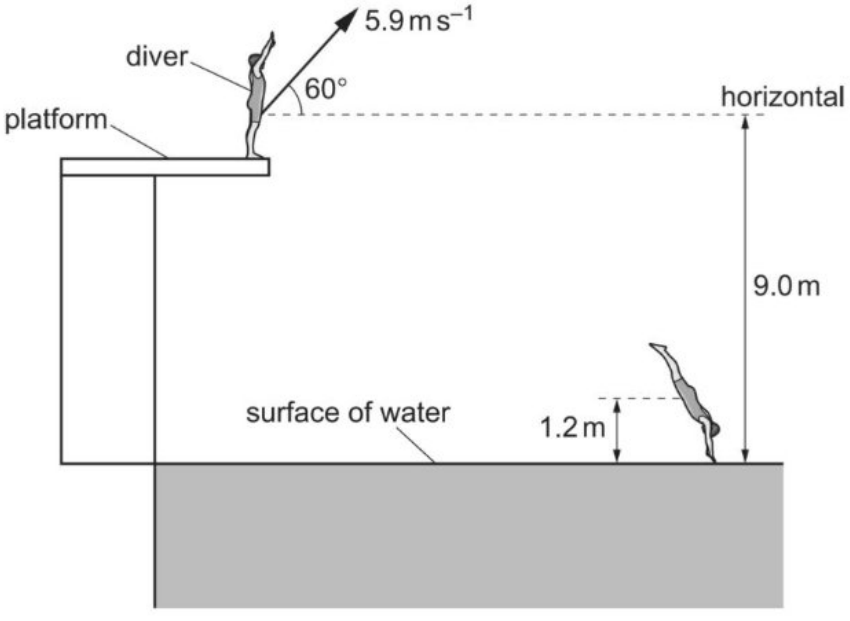


Fig. 1.1 (not to scale)

When the diver is on the platform his centre of gravity is at a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is at a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

speed = ... m s^{-1} [3]

(c) After entering the water, the 78 kg diver experiences an upward drag force of 950 N and a buoyant force of 740 N at a particular instant.

(i) Calculate the magnitude of the resultant force and the acceleration of the diver at this instant.

resultant force = ... N

acceleration = ... m s^{-2} [3]

(ii) Describe and explain the variation of the drag force acting on the diver in the water as he moves downwards.

.....

.....

.....

..... [2]